Experiment 7:
Random Sampling and Sampling Distribution

AIM:

To explore random sampling from a population and understand the concept of sampling distribution
using Python in Jupyter Notebook.

Steps:

1. Generate a Population:
○ Create a population of data with a specified distribution (e.g., normal distribution).
2. Random Sampling:
○ Perform random sampling from the population to create multiple samples of different sizes.
○ Compute sample statistics (mean, standard deviation, etc.) for each sample.
3. Sampling Distribution:
○ Plot histograms or density plots of sample statistics (e.g., sample means).
○ Compare the sampling distribution of the sample statistic (mean) with the population distribution.
4. Central Limit Theorem (Optional):
○ Demonstrate the Central Limit Theorem by showing that as sample size increases, the sampling distribution of the sample mean approaches a normal distribution regardless of the population distribution.

Code:

```
import numpy as np
import matplotlib.pyplot as plt



population_mean = 75
population_std = 15
population_size = 300000
population = np.random.normal(population_mean, population_std,
population_size)



sample_sizes = [20, 40, 80]
num_samples = 1500
```

```
sample_means = {}

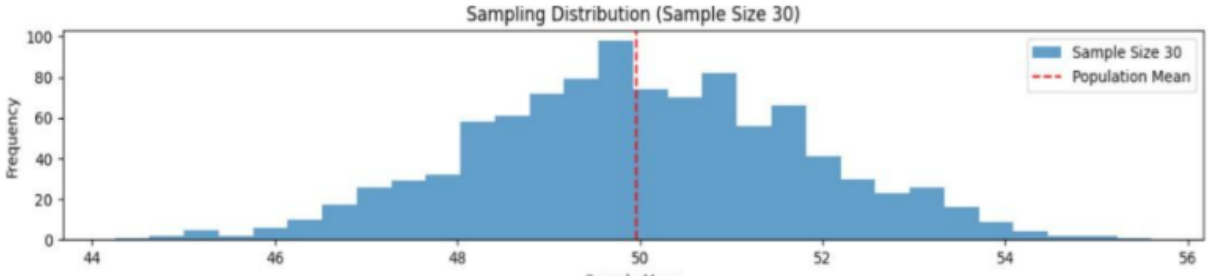
for size in sample_sizes:
    sample_means[size] = []
for _ in range(num_samples):
    sample = np.random.choice(population, size=size,
replace=False)
sample_means[size].append(np.mean(sample))

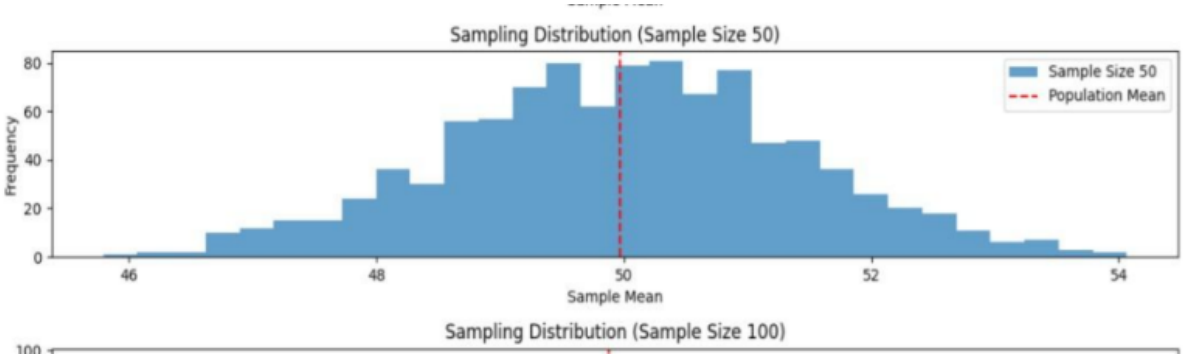
plt.figure(figsize=(12, 8))

for i, size in enumerate(sample_sizes):
  plt.subplot(len(sample_sizes), 1, i+1)
  plt.hist(sample_means[size], bins=25, alpha=0.7,
label=f'Sample Size {size}')
  plt.axvline(np.mean(population), color='red',
linestyle='dashed', linewidth=1.5, label='Population Mean')
  plt.title(f'Sampling Distribution (Sample Size {size})')
  plt.xlabel('Sample Mean')
  plt.ylabel('Frequency')
  plt.legend()

plt.tight_layout()
plt.show()
```

Output:




Sampling Distribution (Sample Size 50)
Sample Size 50
Population Mean
Frequency
Sample Mean
0
20
40
60
80
46
48
50
52
54

Sampling Distribution (Sample Size 100)

Result:

The result show that as the sample size increases the sampling distribution becomes more concentrated around the population mean,demonstration the central limit theorem.